Branching in Git

Data Science Bootcamp

NASHVILLE SOFTWARE SCHOOL

Objectives

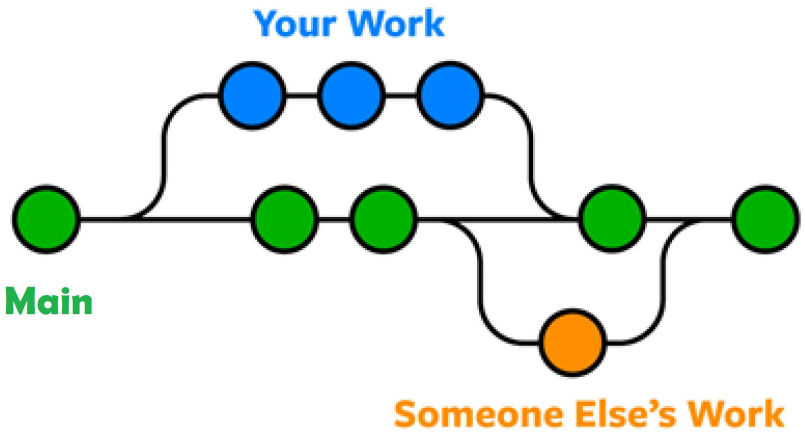
1. Become familiar with the reason branches are used
2. Learn the commands associated with creating, switching to, and pushing to branches in GitHub
3. Understand the commands for the suggested workflow for future NSS team projects

Working in Groups - Code in GitHub

- The original/default branch of any repository is called **main**
- So far, we have only pushed work to GitHub for individual projects, and we've pushed to the main branch (that's why you enter the command "git push **origin main**")
- When a group of people are working together using the same repository, things could get very confusing very quickly if everyone was pushing directly to that repository's main branch
- This is where **branches** come in!
- Branches are created so that each person can continue pushing their work in a way that does not disrupt the main branch
- Typically there will be **one branch for each task** (issue/card) on the **Kanban board**

Branching Out from Main

- Individual branches diverge from the **main** branch so that there is no concern about inadvertently messing with materials stored on the main branch
- In this way, each person can continue on with their work until a given task is finalized and ready to merge back into the main line of development



Branching - Git Commands

- To create a new branch but NOT switch to it:
  - `git branch meaningful_branch_name` (no spaces in the branch name!)
    - e.g. `git branch question_1`
- To see a list of all active branches:
  - `git branch`
  - Note: there will be a star next to the name of the branch you are currently on
- To switch to an existing branch:
  - `git checkout name_of_branch_you_want`
    - e.g. `git checkout question_1`
  - Note: when you switch branches, git changes the files to reflect the last commit **of that branch**
  - You cannot switch branches if you have uncommitted changes
- To create a new branch AND switch to it:
  - `git checkout -b meaningful_branch_name`
    - e.g. `git checkout -b question_1`

Pushing to a Branch - Git Commands

Pushing to a branch follows the same steps as pushing to main, but first you must confirm you are on the branch you want to push to, and you must use the name of that branch in the git push command. Here are the steps:

- `git branch`
  - Confirm there is a star next to the branch name you want to be on
- `git status`
  - Confirm there are no unexpected messages
- `git add .`
  - Remember, the period means to add everything, you can also choose specific files
- `git commit -m "meaningful message about your work"`
- `git push origin relevant_branch_name`
  - e.g. `git push origin question_1`

Recommended Workflow

When working on a group project, these are the steps we recommend you take when setting up to work on GitHub with your team:

- Clone down the team's repository
  - `git clone pasted_link_from_GitHub`
- Create and switch to a branch for yourself in which to work:
  - `git checkout -b meaningful_branch_name`
  - If you don't have a specific/defined task to base your branch name on, you can use your name/initials
  - e.g. `git checkout -b mws_branch`
- Push to your branch as you complete parts of your work